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Article in *Advances in Structural Engineering* · November 2015

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Experimental Study on the Effects of Stacking Sequence on Low Velocity Impact and Quasi-Static Response of Foam Sandwich Composite Structures

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(Received: 19 August 2014; Received revised form: 30 December 2014; Accepted: 7 May 2015)

Abstract: In this study, investigation has been made on the impact response of twill weave carbon fabric/epoxy foam sandwich composites by subjecting two types of stacking sequences of a sandwich composite structure, SC1 and SC2 to quasi-static indentation and low velocity impact loading. SC1 and SC2 had a sequence of [0/90, ± 45 , Core, 0/90, ± 45], [0/90, ± 45 , 0/90, ± 45 , 0/90, Core, 0/90, 0/90, 0/90, 0/90], respectively. This work was done by use of material testing system and an instrumented Drop-Weight Machine (CEAST 9350 drop tower). Foam sandwich composite structures are mainly used in making an Engine Hood. The analysis was done with increase of impact energy on both types of stacking sequences, until complete perforation of the specimens at 25 Joule impact energy occurs. The failure processes of the damaged specimens under the three different impact energies (5 J, 15 J, and 25 J) were evaluated by comparing load–displacement curves. Images of damaged samples were taken from both impacted side and non-impacted side and compared for all impact energies. Cross-sections of damaged specimens were also inspected visually and discussed. The load-displacement curves were obtained to characterize the failure mechanisms in the face sheets and core. Failure modes were also studied by sectioning the samples at the impact location and observed under an optical microscope. The primary damage mode was found due to the fiber fracture, delamination, matrix crack, and foam crack.

Key words: stacking sequences, low velocity impact, quasi-static, fiber failure, perforation.

1. INTRODUCTION

In various engineering fields, composite materials are rapidly replacing the conventional materials due to their specific properties and design flexibility. Besides mechanical properties, control for quality surface finishing of composite parts is most important. Good surface finishing improves not only the aestheticism of the final structure, but also the quality of the part assembled after manufacturing. Several Non-

destructive testing and Experiment (NDT&E) techniques have been studied by Godin *et al.* (2004), Huang *et al.* (2001), Pan *et al.* (2012), and Raišutis *et al.* (2010). Damage to the structural laminates can take the form of matrix cracking, delamination between plies, debonding of the resin fiber interface, and fiber breakages in case of fiber reinforced plastic laminates. While for sandwich laminates, debonding of skins from the core is more common. The amount of energy which

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can be absorbed in a composite structure during structural damage depends on the matrix properties and reinforcement. That includes fabric types, foam properties, fiber orientation, fabrication techniques, and the rate of impact.

The sandwich composite structure panels are being used in aircraft, automotive, and aerospace industries due to their high specific strengths and stiffness. However, sandwich composite structures are susceptible to damage and failure due to transverse contact and impact by foreign objects. According to Andrews and Moussa (2009), Lim and Lee (2004), Raj and Ghosn (2008), and Zenkert and Burman (2011), the failure mode map was found to be a good technique in designing sandwich composite structures. This is to optimize the energy absorption properties and load carrying capacity with minimum weight satisfying the quasi-static and impact loading requirements. Zhu and Chai (2013) studied various modes of damage and failure of composite sandwich panel subjected to quasi-static indentation and low velocity impact. They compared the predicted modes of certain sandwich configurations with the experimental results. The overall predictions were satisfactory, but more work should be done to validate these maps. For impacts that do not give results for complete penetration of the samples, damage consists of delamination, matrix cracking, and fiber failures. Delamination, which is the debonding between adjacent lamina, is of most importance and it reduces the strength of the laminate. Therefore, a lot of studies have been carried out to understand and improve the impact response of composite materials and structures by Abrate (1991), Bibo and Hogg (1996), Cantwell and Morton (1991), Hogg and Bibo (2000), and Richardson and Wisheart (1996).

A new characterization technique to measure and quantify the surface defects in composites based on deflectometry was used by Fotsing *et al.* (2014). Damaged areas can also be investigated visually different techniques such as optical or electron microscopy, ultrasonic B-scanning, C-scanning, and acoustic imaging as reported by Fan *et al.* (2011), and Peters *et al.* (2003). Bull *et al.* (2014) used Synchrotron radiation computed tomography (SRCT) approach for damage assessment and quantification. An approach to evaluate the impact damage initiation and propagation in composite plates was investigated by Luo *et al.* (2001). The presence of internal damage was found by Kim and Sham (2000) to cause substantial degradation in important mechanical properties, including strength and stiffness. Static and low-velocity impact on mechanical behavior of foam sandwiched composites

with different ply angles face sheets was investigated by Mohammed *et al.* (2014). Lightweight composites and sandwich composite structures are susceptible to impact damage, which may severely decrease the structural stiffness, stability, and load carrying capacity. However, the internal damage in the panel during impact is very difficult to inspect, and have been investigated by Abrate (1997), Anderson (2005), Diamanti *et al.* (2005), Feraboli (2006), Feraboli and Kedward (2006), and Lee *et al.* (1993). A new nonlinear high order theory for sandwich beams with analytical and experimental investigations was studied by Dariushi and Sadighi (2014). The experimental results of specimens with different arrangement supported the claims based on analytical predictions about similarities and differences between linear and nonlinear models. In all cases, good agreement was obtained between the nonlinear analytical predictions and experimental results.

An alternative method, energy profiling method (EPM) was used by Aktas *et al.* (2009), based on a variation of the excessive energy (E_e) versus impact energy (E_i), to determine the penetration threshold (P_n). The damage behavior of composite sandwich panels with aramid paper honeycomb (NOMEX) and polyetherimide (PEI) foam cores under transverse impacts at high velocities was investigated by Aktay *et al.* (2005). Low-velocity impact response of composite sandwich beams was studied by an analytical model by Ivañez *et al.* (2014). The predicted results were in good agreement with experimental data in terms of maximum contact force, contact time, and contact force–time curves. But more study needs to be done to validate these maps.

In this study, quasi-static indentation and low-velocity impact responses of two different sandwich composite structures were investigated. Different face sheet stacking sequences of twill weave woven carbon fabrics as face sheet layers were used. The effects of stacking sequence were used to explain the different thicknesses of face sheets caused a difference in bending stiffness of the foam sandwich composites which affected the impact response. The thicker the sandwich structure, the higher the bending stiffness and hence an increased in elastic response and load carrying capability as seen in the results presented in the current study.

2. EXPERIMENTAL MATERIALS

2.1. Materials

ROHACELL® 71 RIST was used as foam core in the sandwich composite structure. The properties of the foam core are presented in Table 1. The skin of the foam sandwich composites used was a twill weaved carbon fabric, 200 g/m² T300/ECC Style 452 3K 2 × 2. Epoxy

Table 1. Properties of the foam material

Properties	Value
Density (kg/m ³)	75
Compressive strength (MPa)	1.7
Tensile strength (MPa)	2.2
Shear strength (MPa)	1.3
Elastic modulus (MPa)	105
Shear modulus (MPa)	42
Strain at break (%)	3
Poisson's ratio	0.3

resin, EPIKOTETM MGS® RIMR135 and hardener, RIMH 137, were used with the ratio of 100:30 by weight. The calculated mechanical properties of the skin are presented in Table 2 and the following equations were used to calculate the mechanical properties of the lamina:

$$E_1 = E_F V_F + E_M V_M \quad (1)$$

$$\frac{1}{E_2} = \frac{V_F}{E_F} + \frac{V_M}{E_M} \quad (2)$$

$$\frac{1}{G_{12}} = \frac{V_F}{G_F} + \frac{V_M}{G_M} \quad (3)$$

$$\nu_{12} = V_f \nu_f + V_m \nu_m \quad (4)$$

$$\rho_c = \rho_f V_f + \rho_m V_m \quad (5)$$

where, E_1 is the longitudinal stiffness, E_F and E_M are the respective stiffness of fiber and matrix, V_F and V_M are the respective volume fractions of fiber and matrix. E_2 is the transverse stiffness, G_{12} is the shear modulus, G_F

Table 2. Mechanical properties of the face sheet

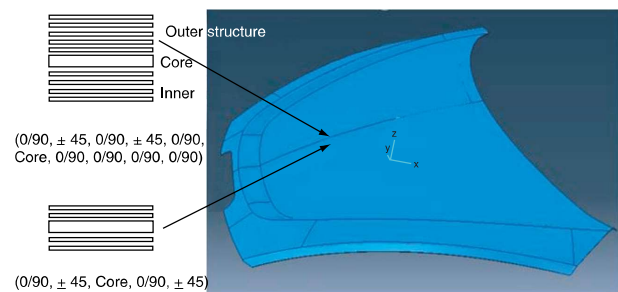
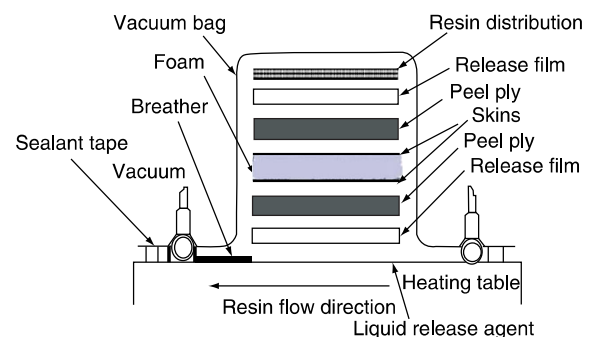
Properties	Value
Longitudinal stiffness	$E_1 = 139000$ MPa
Transverse stiffness	$E_2 = 7330$ MPa
Shear modulus	$G_{12} = 29$ MPa
Poisson ratio	$\nu_{12} = 0.26$
Density	1500 kg/m ³

and G_M are the respective shear moduli of fiber and matrix, ν_f and ν_m are the Poisson ratios of fiber and matrix respectively while ρ_f , ρ_m and ρ_c are the respective fiber, matrix, and composite densities.

The foam sandwich composites have two different stacking sequence face sheets: SC1, SC2 of stacking sequence [0/90, ±45, Core, 0/90, ±45], [0/90, ±45, 0/90, ±45, 0/90, Core, 0/90, 0/90, 0/90, 0/90] respectively. A setup of stacking sequences employed in the engine hood is shown in Figure 1. The total thickness of the foam sandwich composite structure was approximately 12-13 mm including the upper skin, foam, and lower skin. A setup of Vacuum Assisted Resin Infusion (VARI) is shown in Figure 2. Composite sandwiches were retained at a constant pressure of 0.1 MPa and a temperature of 60°C for 6 hours for curing.

2.2. Testing methods

The fabricated foam sandwich composites were of two different stacking sequences face sheets. The upper and lower skin consisted of different ply numbers depending on the type of stacking sequences. Scanning electron microscopy was used to check the defects on the surface of the composite specimens by using a varied magnification scale, such as x60, x100, x500, and x1000, as shown in Figure 3.

**Figure 1.** Engine hood stacking sequences**Figure 2.** Vacuum assisted resin infusion

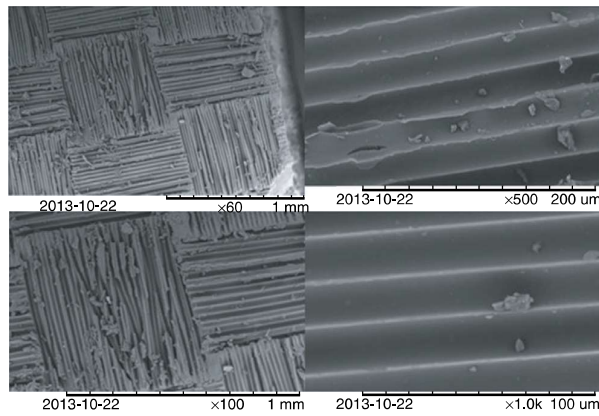


Figure 3. SEM images of composite sandwich specimens

2.2.1. Quasi-static indentation

Material test system, MTS 810® was used for quasi-static three-point bending test, as shown in Figure 4. Loading was done at a speed of 2 mm/min, and the indenter diameter was 12.7 mm. The specimen dimensions are listed in Table 3.

2.2.2. Low -velocity impact

Instron CEA9350® was used to conduct the low-velocity impact testing of foam sandwich composite structures (Figure 5). The test was performed by

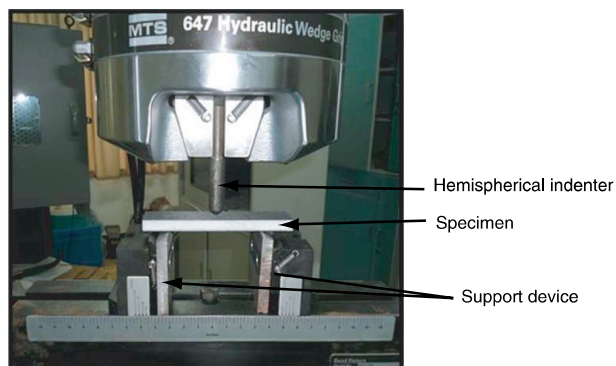


Figure 4. Material Test System machine



Figure 5. CEA9350 impact tester

dropping a 16 mm diameter hemispherical striker carrying a total weight of 5.277 kg on the foam sandwich composites specimens from various heights. The total weight included a carriage weight of 4.3 kg, a striker weight of 0.63 kg, and 0.347 kg additional weights. Three levels of energy were applied to the foam sandwich composites specimens: 5 J, 15 J, and 25 J. The panel size was 150 mm x 100 mm, as shown in Figure 6. The incident (initial) velocities for the three levels were 1.38, 2.38, and 3.08 m/s corresponding with drop heights of 97.09, 289.78, and 483.044 mm, respectively. The impact test was done according to ASTM 7136. At least three samples were subjected to testing at each impact energy level. In addition, the load-displacement histories of impact energy against absorbed energy, time, and peak load for all the samples were recorded.

Table 3. Dimensions of the specimens which were used in the quasi-static 3-point bending tests

Sample number	Lay-up	Total length (mm)	Span length (mm)	Width (mm)	Core thickness (mm)	Skin thickness (mm)
1-1	SC1	135	80	30	10	1
2-1	SC1	135	80	30	10	1
3-1	SC1	135	80	30	10	1
1-2	SC2	135	80	30	10	1.5
2-2	SC2	135	80	30	10	1.5
3-2	SC2	135	80	30	10	1.5



Figure 6. Composite sandwich panel

3. RESULTS AND DISCUSSIONS

3.1. Quasi-static test

3.1.1. SC1[0/90, ±45, Core, 0/90, ±45]

This section describes the experimental results after the SC1 type of foam sandwich composites specimens that were tested under quasi-static three point tests. All results were plotted in terms of applied load versus displacement. All the specimens for the two stacking sequences were of the same span length, thus making it possible to compare their load/displacement curves. This will allow more accurate comparison of the resulting curves. Figure 7 explained the response behavior of foam sandwich composites of SC1 to quasi-static loads. The curve exhibited a brittle behavior which means a little energy was required to initiate a crack, and then a little more to propagate it to the final failure. The behavior of the foam sandwich composites was bending stiffness until the peak load since no

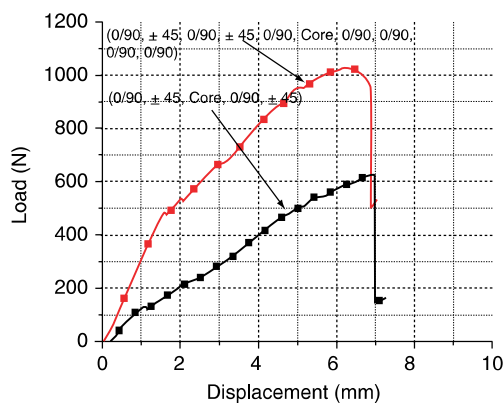


Figure 7. Load-displacement curves for quasi-static tests of sandwich composites

delamination was observed. There was no oscillation before peak load which resulted from vibrations on the supports, no defects in foam sandwich composites, and a little initiation of damage in the specimens. After the load reached a peak value, significant drop of about 30–60 % of the peak load in the load–displacement curve was observed. This sudden drop was due to the fiber cracking and foam crack. In this case, plateau region was not observed until the final failure. After the load drop, the specimen did not sustain extra load because the foam sandwich composites reached a critical point.

3.1.2. SC2 [0/90, ±45, 0/90, ±45, 0/90, Core, [0/90]₄]

The load–displacement curves (Figure 7) were used to determine the behavior of the failure modes of the foam sandwich composites with different stacking sequence of the face sheets. The curves can be divided into two regions; the first region observed the bending stiffness of the foam sandwich composites, which is a linear behavior in appearance corresponding to an elastic deformation of the foam sandwich composites. The deformation represents when the displacement of the indenter was affected in the upper surface (compressive side), due to the presence of some imperfections on the surface of the foam sandwich composites. From the end point, bending stiffness of the curve showed nonlinear behavior attributed to the combined effect of material compressive yielding, delamination, debonding, and geometric nonlinearity. The second region represents when the load reached the peak value, a significant drop of about 20–50 % of the peak load in the load–displacement curve is observed. This sudden drop was due to the fiber cracking, delamination, and foam crack. Here the plateau region which was an indication of the penetration through the foam core, but the indenter was not observed. This means the indenter did not penetrate the foam. The final failure occurred when the upper skins were crushed due to the compressive failure. In some cases, the skin and the foam debonded due to core shear failure.

From Figure 7, by comparing the peak loads of the foam sandwich composites in quasi-static test, the SC2 stacking sequence has a higher peak load than the SC1 stacking sequence. This variation in peak load can be attributed to the fiber orientation, reinforcement performance, amount of materials involved in both samples, and the total number of the layers.

In Figure 7, the linear behavior of the SC1 stacking sequence was observed that shows the load carried by the foam sandwich structure initially increased while the core and the skins were being stressed. The specimens

with this stacking sequence showed the lowest load among the tested samples, a high displacement; indicating an elastic property. 'Mixing of the curves' behavior was observed in SC2 stacking sequence, where the first stage showed a linear behavior, second stage presented the damage initiation and progression. This consists of delamination in the compressive side, debonding and foam crack until the peak load, after which there was a failure of the foam sandwich structure. The specimens with this stacking sequence showed the lowest displacement and highest peak load, resulting into a ductility behavior that was observed on the slope of the load-displacement curves.

Structural stiffness of this foam sandwich structure depended on the strengths of the outer skins, foam and the interfacial bond strength between the skins and core. Changing the stacking sequence clearly showed differences in the resulting properties under the quasi-static test of the foam sandwich structures.

3.1.3. Failure mode mechanism

The damage in the face sheets subjected to static loading was observed due to matrix cracking, delamination, and fiber breakage. Quasi-static tests of the foam sandwich composites beams were performed to investigate the failure loads and modes. The static failure modes of sandwich beams are affected by the geometries of the specimens, boundary conditions, and material properties. The unexpected failure of the specimen during quasi-static as indicated earlier, occurred due to

the load transferred to the core and was more than its strength during a skin failure. Since core materials generally had lower mechanical properties than the skins (due to their lower density), the damage initiation and the local indentation behavior of the sandwich beams was affected which led to the core failure before the face sheet. The experimental investigation showed that under static loading, the specimens failed with sudden brittle type failure in SC1. This was due to shear failure of the core and compressive failure of the skins. By comparing the failure modes of the foam sandwich composites in the two types of sandwich composite stacking sequences, it was observed that it had no similar failure behavior.

Figure 8(a) represents specimen before the test and Figures 8 (b), (c), and (d) show during the test. Foam crack in Figure 8(c) and (d), while upper skin crushing failure [Figure 8(b)] appeared as a result of compression. Tensile strength in the upper skin was observed in both types of stacking sequences. Multiple cracks were generated along the tensile side of the specimen with all the cracks propagating toward the compressive side. However, the final failure occurred when the upper skin crashed (compressive side), and the skin-foam underwent debonding. During this process of crack propagation, debonding between the foam and bottom skin was observed only in SC1. Indentation of the face sheet occurred at the indenter location as a result of the fiber breakage. The exact order and the relative size of these failure modes depended on the face sheet configuration.

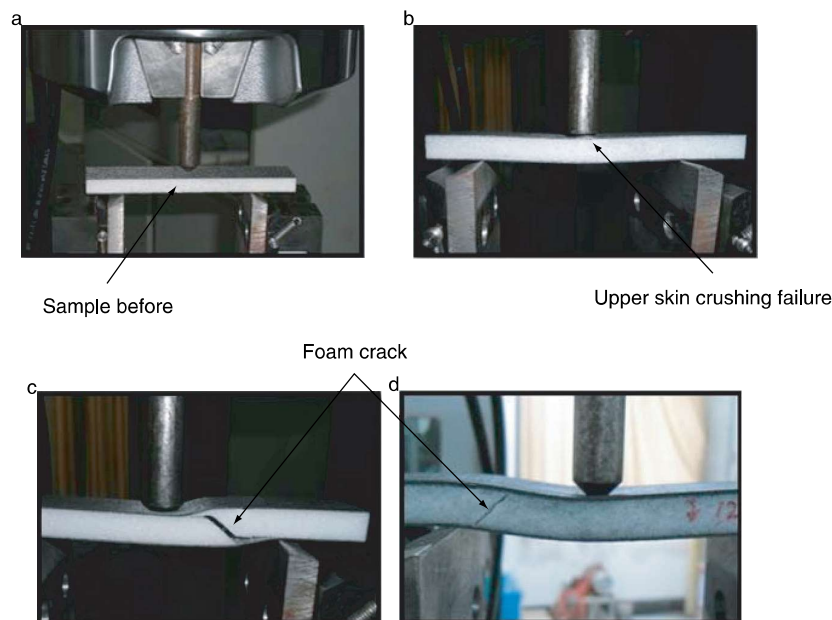


Figure 8. Failures modes of sandwich structure composite

Table 4. Damage area and delamination measurement

Sample number	Lay-up	Layers number	Structure type	Core thickness (mm)	Skin thickness (mm)	Delamination length (mm)	Area (mm ²)
1-1	SC1	4	Sandwich	10	1	4.26	2.43
2-1	SC1	4	Sandwich	10	1	4.36	1.4
3-1	SC1	4	Sandwich	10	1	4.30	1.73
1-2	SC2	9	Sandwich	10	1.5	–	0.837
2-2	SC2	9	Sandwich	10	1.5	–	1.00
3-2	SC2	9	Sandwich	10	1.5	–	1.6

Table 4 explains the measurement of damage area and delamination of the sandwich structure. It can be observed that there is no delamination through the thickness of SC2 stacking sequence due to different fiber orientation. Using visual inspection and optical microscopy (Figures 9 and 10), the comparison of the damage area showed that the SC1 had a larger size of the damage area and delamination length through the thickness than SC2. This can be attributed to the fiber orientations, material amount involved in both structures, and the total number of layers in the upper and lower skins. This test was conducted using different stacking sequences of the foam sandwich composites to be able to get a variation in the failure modes.

3.2. Impact test

The impact behavior of the foam sandwich composites with different stacking sequences of the face sheets was by using the drop weight testing method with different impact energy. A number of tests were performed under three impact energies (5 J, 15 J, and 25 J) in order to

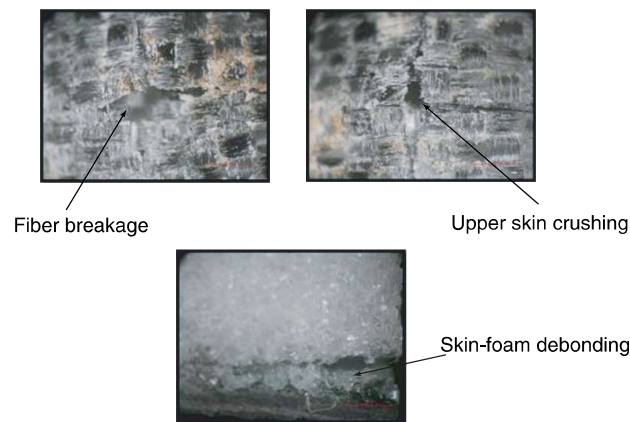


Figure 9. Optical microscopic pictures of [0/90, ±45, 0/90, ±45] for damage area and through the thickness

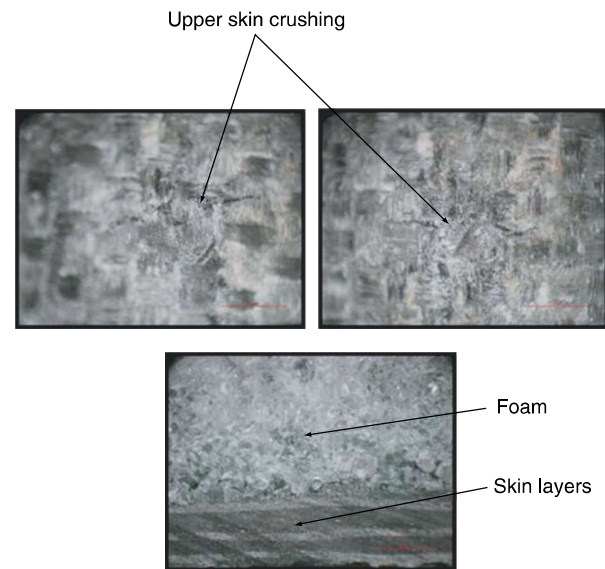


Figure 10. Optical microscopic pictures of [0/90, ±45, 0/90, ±45, 0/90, 0/90, 0/90, 0/90] for damage area and through the thickness

examine the damage process in the different stacking sequences; SC1 and SC2. The overall damage expansion in specimens after an impact event was evaluated by visual inspection. The failure modes involved in impact damage under varied impact energies were characterized by the upper skin failure, core shear failure, lower skin failure, and penetration depth. A characteristic of load–displacement curves also included some useful tips in assessing the damage process of the composite sandwich structures.

Absorbed energy in an impact event can be calculated from the load–displacement curves as shown in Table 5. The following equation was used to calculate the absorbed energy, according to Kenny and Torre (1998); and Torre *et al.* (1997):

Table 5. First peak loads of foam sandwich structure with displacement for the different levels of the impact energy.

Sandwich structure lay-up	Impact energy (J)	Absorbed energy (J)	First peak load (N)	Displacement (mm)
SC1	5	4.5	658	3.5
	15	16.2	807.7	4
	25	24.9	848.7	7.23
SC2	5	3.94	1805.3	3.85
	15	15.25	1739.3	4.51
	25	25.7	1750.5	4.26

$$E_{ab} = \frac{mv^2}{2} - \frac{mv_{i(t)}^2}{2} \quad (6)$$

Whereby, E_{ab} = Absorbed energy, v = an initial velocity, $v_{i(t)}$ = velocity of the impactor at the time and m is the mass of the striker.

3.3. Performance Index

In order to develop a reference scale for the evaluation of crash performance of the tested structures, some non-dimensional parameters were introduced. The information from the impact tests was used to calculate the absorbing energy moment parameter (AEMP) and performance index (PI) respectively. In order to measure the maximum energy that the structure can absorb with respect to the moment produced due to impact load, the absorbing energy moment parameter (AEMP) was calculated at various impact velocities (1.38, 2.38, and 3.08 m/s).

$$AEMP = \frac{E_{\max}}{M_{\max}} \quad (7)$$

where,

$$M_{\max} = \frac{P_{\max} \times a}{4} \quad (8)$$

The Performance Index;

$$PI = \frac{AEMP}{RD^2} \quad (9)$$

where,

$$RD = \frac{Def_{\max}}{a} \quad (10)$$

‘ a ’ is the span length (mm) of the impact specimen, ‘ $E_{\max}$ ’ is maximum absorbed energy (J), ‘ $M_{\max}$ ’ is a

maximum moment in the panel corresponding to the maximum contact force, ‘ $P_{\max}$ ’ is maximum contact force (N), ‘ $Def_{\max}$ ’ is the maximum deformation (mm).

AEMP is the ratio of the maximum energy absorbed by the plate ($E_{\max}$) to the maximum moment ($M_{\max}$) of the impact load on the supports. The performance parameter (PI), which essentially relates to the impact energy absorbed to the deflection, is the ratio of the AEMP to RD.

Eqns 7, 9, and 10 explain the non-dimensional parameters which were introduced to evaluate the crash performances of the panel. In particular, the deformation ratio RD (which is the maximum panel deformation divided by the span length) was used to compare the deformation due to the impact on panels of different length. The ratio of the maximum energy absorbed to the maximum moment of the impact load on the supports (AEMP) in Eqn 7 is a parameter proposed to measure the capability of a structure to absorb energy. It relates to the impact energy to the load transferred on the supports. By dividing AEMP by RD^2 , it is possible to obtain another non-dimensional parameter PI represented by Eqn 9, which gives more precise information on the crash performance. Since this parameter relates the energy absorbed by the structure to the non-dimensional deformation, it can give more complete information about the crash performances of the panels. The results of the AEMP ratio and PI value are presented in Table 6.

From the results in Table 6, a high PI value was observed in SC1 stacking sequence at 5 J which indicates that the sandwich plate was able to absorb high amounts of energy without transferring excessive deformation to the inner structure. This is one of the main requirements of a crash resistant structure for transportation. A high AEMP ratio was observed in SC2 stacking sequence at 25 J which indicated a lower amount of load was transferred to the supports as compared to the impact energy. A high AEMP ratio was also observed at the same level of 25 J impact energy in both of the stacking sequences at 3.078 m/s. This

Table 6. Summary of the results of non-dimensional parameters

Lay-up	Velocity (m/s)	Impact energy (J)	Absorbed energy (J)	High peak load (N)	AEMP	PI
SC1	1.38	5	4.5	825.2	0.0032	0.44
	2.384	15	16.2	927.8	0.0013	0.0084
	3.078	25	24.9	885.8	0.007	0.014
SC2	1.38	5	3.94	1915.3	0.0006	0.32
	2.384	15	15.25	1797.3	0.0011	0.06
	3.078	25	25.7	1985.2	0.013	0.25

showed that specimens exhibiting high AEMP when impacted at 3.078 m/s, transferred less amount of energy to the base. High *PI* values were observed in the same level of the impact energy (5 J at 1.38 m/s) which indicated that the structure has a high crash resistance transferring very little deformation to the base layers.

Figures 11 and 12 illustrate the load-displacement plots for foam sandwich composites of SC1 and SC2 at 5 J, 15 J, and 25 J respectively. The pictures of the upper and lower skins and cross-sectional view for the foam sandwich composites with SC1 and SC2 are presented in Figures 13 and 14. At all the impact energy levels within the foam sandwich composites, the peak load was between 0.8 and 2 kN

Analyzing the first peak load in the curves in Figures 11 and 12, the load was found to increase with increasing impact energy levels as shown in Table 5. In this area of the first peak load, the foam sandwich structure has a linear behavior due to the resistance of the composites to carry further load up to a certain point. This depended on a composite lay-up, the interface bonding between the skins, core and the material amount involved in both

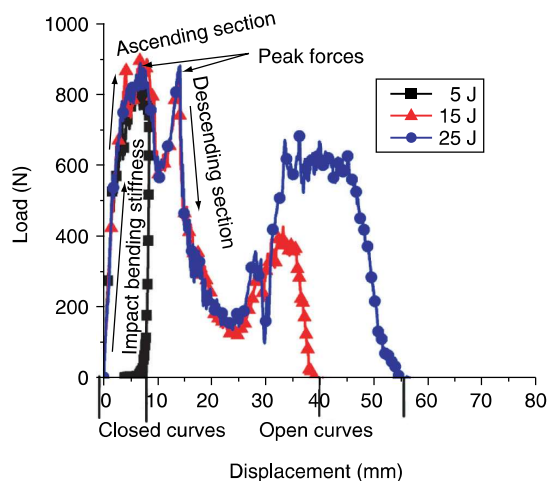


Figure 11. Load-displacement curves for low-velocity impact tests of sandwich composites with [0/90, ±45, 0/90, ±45] at 5, 15, and 25 J

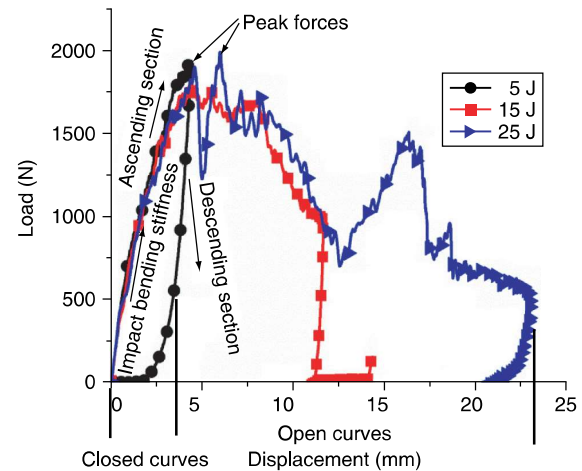


Figure 12. Load-displacement curves for low-velocity impact tests of sandwich composites with [0/90, ±45, 0/90, ±45, 0/90, 0/90, 0/90, 0/90] at 5, 15, and 25 J

structures. The variation of displacement from low to high impact energy was referred as the penetration depth of the indenter through the three facings of the foam sandwich composites (top face sheet, foam, and bottom face sheet). The load-displacement curves were also showing a similar tendency; exhibiting a linear increase with a small deformation. This behavior was attributed to the condition when the indenter comes in contact with the specimen, and initiated delamination in the top face sheet. The global shape of the damage observed was approximately circular while individual ply damage exhibited a global type of damaged area.

Figure 11 showed the typical load-displacement curves of SC1 at different levels of impact energy. Individually, each curve has an ascending section of loading reaching to a maximum load value and a descending section of unloading at 5 J in this situation. The ascending section of the load-displacement curve is known as bending stiffness due to the resistance of the composite to impact loading. At this section also, the maximum load reached the highest value, known as the

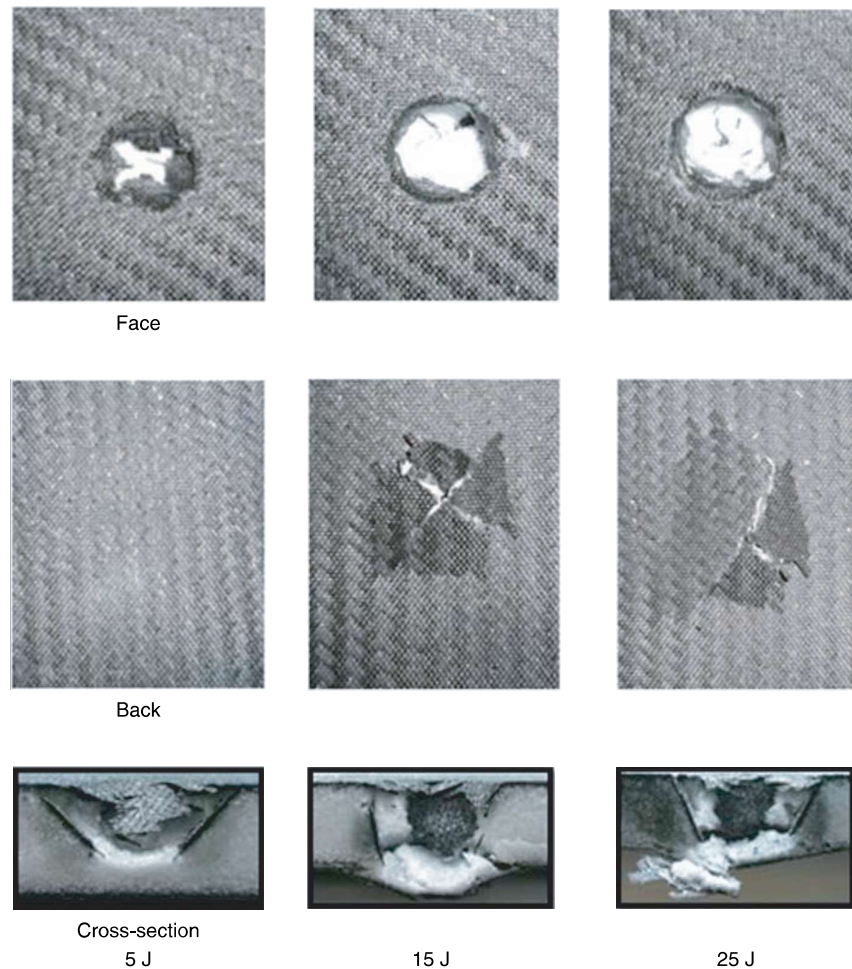


Figure 13. Pictures of the top, bottom surfaces and cross-sectional view of $[0/90, \pm 45, 0/90, \pm 45]$ at 5, 15, and 25 J

peak force. Typical load-displacement curves of the foam sandwich composites subjected to impact loading can have three situations; rebounding, penetration, or perforation.

Load-displacement curves in general, can be classified as; closed type curves and open type curves. The rebounding case results in closed curves indicating the rebounding of the impactor from the specimen surface. The closed type curves return back to the origin of the graph after the descending section from the peak force value as shown in two types of the stacking sequences only under 5J impact energy level. This is due to the mechanical behavior of the specimen tends to have elastic response and resistance to impact loads (Figures 11 and 12). The area under the load-displacement is the absorbed energy.

When the impact energy increases, the closed type curve bound a larger area and displacement increases while rebounding section becomes smaller. As the impact energy continues to increase, the curve type

changes from the closed type to the open one. If a curve is of an open type, the specimen is either damaged or perforated by the impactor. Two key indices have been identified in the area of foreign body impact on composite sandwich structures; damage resistance and damage tolerance. Damage resistance is a measure of the ability of sandwich composite structures to resist damage when subjected to foreign body impact and damage tolerance is a measure of the ability of sandwich structures to perform satisfactorily with impact damage and without any perforation. This is explained in the pictures of the sample at 5 J in Figures 13 and 14. It has been observed that the damage was limited, i.e. just at the top face sheets without further propagation to the foam and the bottom face sheet. The size of the damage in SC1 at 5J was higher than SC2 due to fiber orientation and total thickness of the foam sandwich composites. A sudden drop of the peak load, in the load-displacement curves at the impact energy level (15 J) implies that the indenter penetrated the top face sheet through the foam

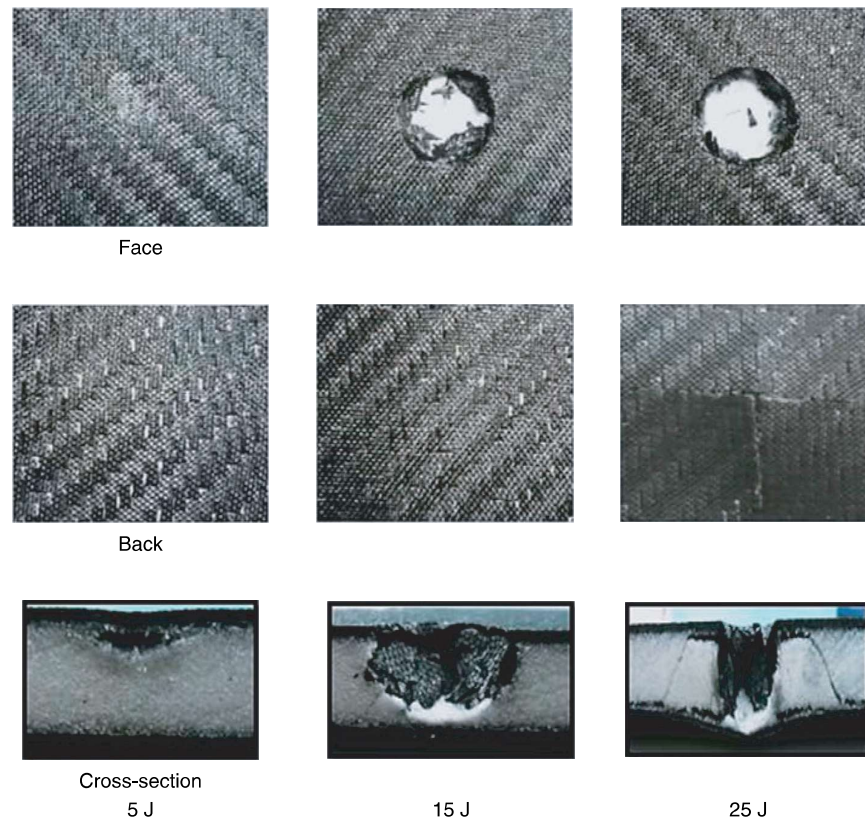


Figure 14. Pictures of the top, bottom surfaces and cross-sectional view of $[0/90, \pm 45, 0/90, \pm 45, 0/90, 0/90, 0/90, 0/90, 0/90, 0/90]$ at 5, 15, and 25 J

and has come in contact with the bottom face sheet, Figure 13. In Figure 14, it is observed that the indenter penetrates the top face sheet and comes in contact with the foam without contact to the bottom face. While in Figure 13, the indenter penetrated the upper face sheet, through the foam and makes contact with the bottom face sheet. This is attributed to fiber orientation, upper face and bottom face sheet thickness. At 15 J, it is noted that, complete failure of the top face sheet exposes the foam leading to penetration (Figures 13 and 14) and contact with the lower face sheet and cracking, as shown in Figure 14. When the impact energy increased, the displacement also increased, resulting in a high damage area, delamination, matrix crack, fiber breakage, foam crack. This also led to the indenter penetrating the foam and making contact with the bottom face sheets. A secondary peak value was also observed, and it was attributed to the load carried by the bottom face sheets. After the second peak, many oscillations were observed; the cause was attributed to dents and delamination in the top and bottom face sheets.

For 25J energy level, the specimens were completely penetrated followed by the final failure. This is quite evident in the picture of samples impacted at 25 J as

shown in Figures 13 and 14. The top face sheets and foam were penetrated, and the bottom face sheets remain without penetration but with some damage. The high-impact energy level 25 J shows different behavior in the load-displacement curve because of the effect of stacking sequences used, critical point of the high load, and fiber orientation. In summary, five failure modes were observed at three levels of impact energy; 5 J, 15 J, and 25 J which were delamination, matrix crack, fiber failure, foam crack, and penetration mode. Delamination is a separation of plies which progresses in the resin rich area between the plies and it is as a result of the mismatched bending stiffness between adjacent layers. The delamination area which was observed in the foam sandwich composite specimens was generally oblong shaped with its major axis being coincident with the fiber orientation of the layer at the interface.

Matrix damage was induced by transverse low velocity impact, and took the form of matrix cracking, fiber/matrix debonding, and delamination initiation. Barely visible or minimal damage occurred at low impact energy levels as in 5 J. Matrix cracks are usually oriented in planes parallel to the fiber direction. The type of matrix cracking was dependent

on the global characteristics of the impacted specimens. Fiber failure occurs later than matrix cracking and delamination in the fracture process. It occurred just below the striker due to local high stresses, and indentation effects in the non-impacted face, which was attributed to the high bending stresses. This was clearly observed within both upper and bottom face sheets at 15 and 25 J (Figure 13). However, it was only observed in the upper face sheet at 15 J and in both upper and lower face sheets at 25 J (Figure 14). At 5 J it only occurred in the face sheet for cases, but damage area was bigger in SC1 than SC2 specimens. The final failure occurred when the upper skins were crushed due to compressive failure or in some cases, the skin and the foam debonding resulting from the core shear failure. Finally, the penetration was in macroscopic mode and occurred when the fiber failure reached a critical extent, enabling the impactor to completely penetrate the specimens. The impact energy penetration threshold increased rapidly with specimen thickness. Other various factors, including; fiber sizing, orientation, weaves architecture, matrix type, and interface had an influence on the penetration process [11]. A total of 18 samples was tested under low velocity impact conditions at 5 J, 15 J, and 25 J. Three samples of two different sets of sandwich stacking sequences were chosen for this test. In general, the peak load did not

vary much within same sandwich structures at different impact energy levels. For the same impact energy, the samples of SC2 showed high peak load than SC1. The energy absorbed by elastic deformation of the foam sandwich composite was calculated from the area under the load-displacement curve and this increased with the increase in the impact energy levels for all sandwich structures. At low and medium energy levels, all energy was absorbed by the samples, and thus the samples did not fail completely. In the case of high-impact energy levels, the absorbed energy was higher because the sandwich composites failed. Higher-energy absorption levels were caused by the significant damage that occurred in three faces of foam sandwich composites.

3.3.1. Failure modes mechanism

When a foreign object impacts on a foam sandwich composite, several damage modes including: delamination, fiber breakage, and matrix cracks occurred. In general, the dominant damage mode depends on, the impact parameters of the impactor and the material properties of the composites. With careful examination of the specimens, in this study, the primary damage mode was observed as the fiber breakage followed by delamination, matrix cracks, foam crack, and penetration mode. Figures 15 and 16 showed the top view, bottom surface and cross-sectional area of the

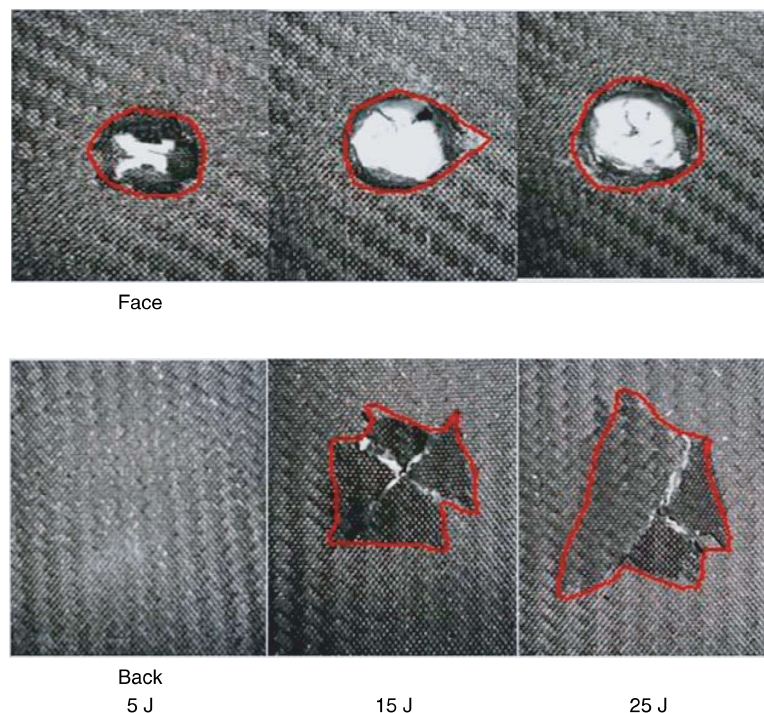


Figure 15. Damage modes on impact face sheets and bottom face sheets of [0/90, ±45, 0/90, ±45]

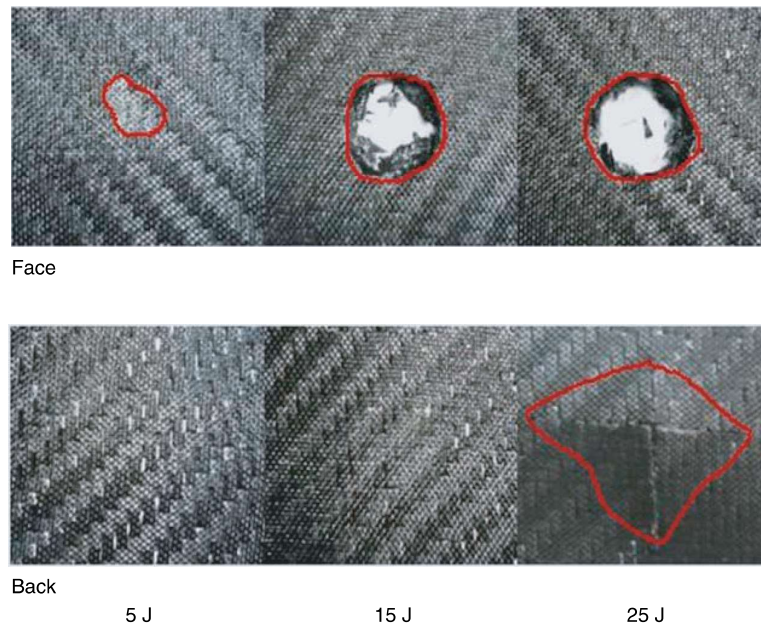


Figure 16. Damage modes on impact face sheets and bottom face sheets of $[0/90, \pm 45, 0/90, \pm 45, 0/90, 0/90, 0/90, 0/90, 0/90]$

foam sandwich composites of SC1 and SC2 specimens. For lower impact energies (less than approx. 25 J), the main damage mode was detected as delamination and matrix cracks rather than fiber fracture. However, for the higher impact energies, splitting between fiber and matrix and fiber fractures were the dominant damage modes around the point of impact. A schematic illustration representing these damage modes is shown in Figures 15 and 16. From the Figures, a back side (non-impacted) view for damaged specimens is depicted. From 5 - 25 J (Figure 15), the damage area is observed that have the same shape of the damage, but the size of damage area increased with the increase in the impact energy at face sheet layers. For this situation, the specimens had penetration in two levels of the impact energy as in 15 and 25 J due to the fiber failure reaching a critical extent, enabling the impactor to completely penetrate the specimens.

From Figure 16, a small damage area was observed at 5 J, when compared with the same level of impact energy as in Figure 15 due to differences in stacking sequences leading to different damage modes. The damage area was observed to have the same shape at 15 and 25 J, though the size of damage area increased with the increase in impact energy at the face sheet layers. As shown in Figure 14, at 15 Joules energy level, when the impactor came in contact with the face sheets without penetration, debonding occurred between the face sheets and the foam. While at 25 J,

the penetration mode occurred in specimens with fiber breakage and delamination.

The top face sheets suffered significant damage in the form of fiber breakage, however, no delamination was observed at the interface between the plies. The reason for this was that all the adjacent plies had different fiber orientation and this behavior was found within all impact energy levels. It shows the greatest damage size in the top and bottom face sheets, which could be attributed to the fiber orientation in the direction of the major axis which caused the greatest splitting in the fiber direction. The manufacturing defects in the laminates played an important role in the failure mode of the foam sandwich composites. The defects like non-uniform distribution of the resin in the samples, poor adhesion between the overall layers of the composite and the void defects could have caused delamination and debonding under impact and load conditions. They also affected the mechanical behavior of the foam sandwich composites such as the stiffness and strength. The delamination shapes, Figures 15 and 16, could be generalized as circular and this behavior was attributed due to the shape of the indenter and a combination of the fiber orientation of the foam sandwich composites. Table 7 indicated the damage parameters for foam sandwich composites at 5, 15, and 25 J energy levels. Table 7 and Figures 13 and 14 shows that the penetration depth became deeper and the maximal cracking width bigger with increasing impact energy levels.

Table 7. Damage parameters for sandwich composite structure with different stacking sequences

Lay-up	Impact energy	damage area (mm ²)	Penetration depth (mm)
SC1	5	201.4	8
	15	281.2	15
	25	375	19
SC2	5	68.6	1
	15	223.7	10
	25	253.9	16

A cross-sectional observation of the specimens, (Figures 13 and 14), showed that the SC1 stacking sequence had the highest penetration depth of up to 19 mm at 25 J as compared to other samples being used at the same level of the impact energy. At high and medium impact energies, the three faces of SC1 specimens were completely penetrated. However, at 5 J the bottom face sheet was not penetrated in both types of stacking sequences used. The specimens of SC2 had the lowest penetration depth of up to 16 mm at the 25 J level of impact energy, and complete penetration of the three faces only took place at the 25 J level of impact energy. The cracking width was found to increase with the increasing impact energy for all the samples. In general, for either stacking

sequences, the extent of the overall damage area and fiber failures increased significantly with the increase in the impact energy.

Figures 17 and 18 explains the impact energy against damage area curves with the shape of the damage which occurred in face sheets for SC1 and SC2 specimens, respectively. It was observed that the damage area (delaminated area) increased with an increase in impact energy. Comparing the peak loads of the sandwich composite structures in impact test, it is evident that the SC2 had a higher peak load compared with SC1 as shown in Figure 18. This variation in peak load can be attributed to the fiber orientation, reinforcement preform, and total number of the layers, and thickness of skin face sheets.

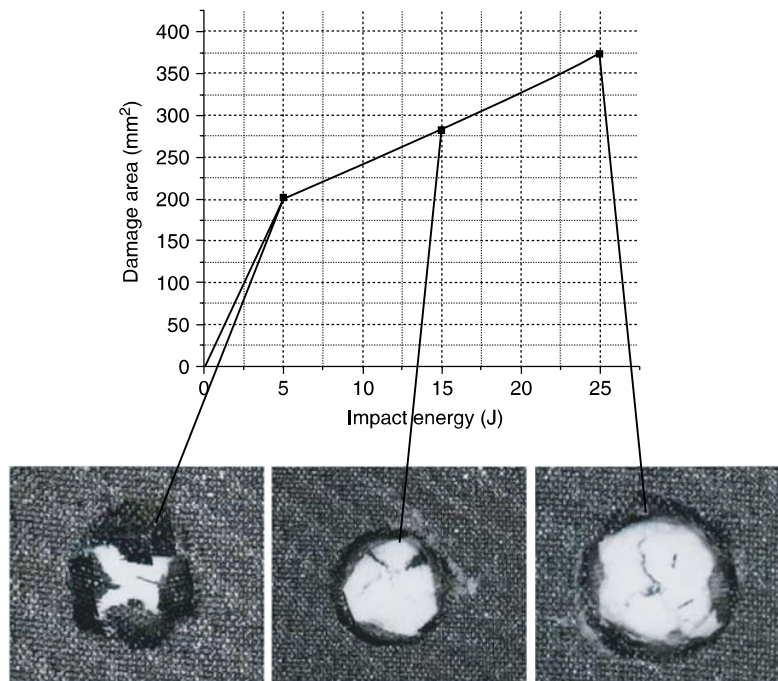


Figure 17. Damage area of sandwich composite structure of [0/90, ±45, 0/90, ±45] versus impact energy after impact

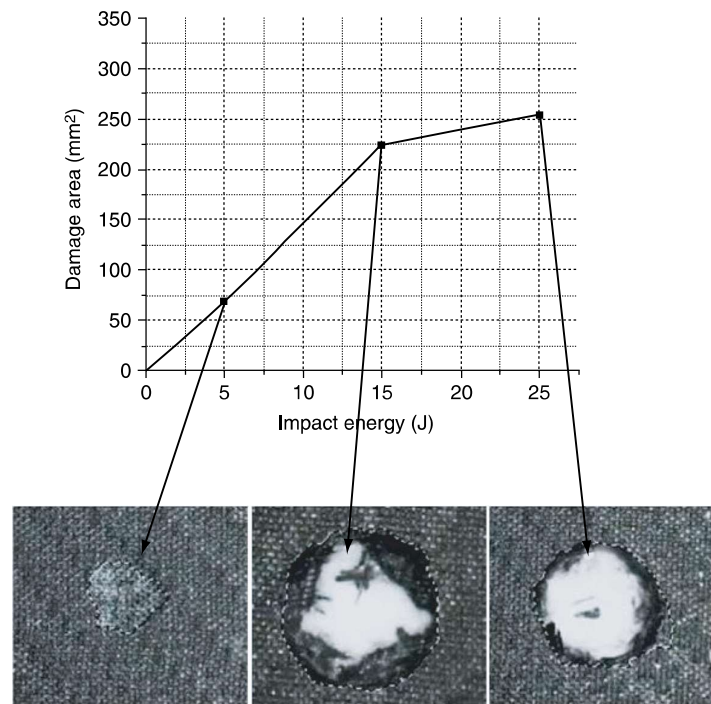


Figure 18. Damage area of sandwich composite structure of $[0/90, \pm 45, 0/90, \pm 45, 0/90, 0/90, 0/90, 0/90, 0/90]$ versus impact energy after impact

4. CONCLUSIONS

The study deals with the investigation of quasi-static, low velocity impact loading, and the damage mechanism process of foam sandwich composites. The following conclusions can be drawn from the tests:

- In the quasi-static test, the damage modes detected by optical microscopy in SC1 specimens were fiber crushing in the upper skin, fiber breakage, matrix crack and delamination between layers in the thickness direction. It was also observed to have a higher damage area due to its fiber orientation and thickness of the face sheets skins.
- Damage modes in SC2 specimens were found to be a small delaminated area due to fiber orientation and foam crack due to the upper skin experiencing a more elastic deformation and reaching a critical point in carrying the load.
- As summarized in quasi-static test, the effect of stacking sequences and properties of the reinforcement and matrix played an important role in the response of the foam sandwich composites. SC2 specimens carried a higher load while the SC1 specimens carried a lower load. It was also evident that varying the stacking sequences of foam sandwich composites changed the failure mode behavior.
- In low velocity impact tests, the results showed that the SC1 specimens had the highest penetration depth of up to 19 mm at 25 J as compared to other samples at the same level of the impact energy. Also the penetration depth increased with an increase in the impact energy.
- At the 5 J level of the impact energy, it was observed that for both types of stacking sequences, the impactor rebounded without penetrating the specimens due to the stiffness of the face sheets.
- The damage area was observed that have shape of damage at 15 and 25 J for both types of the stacking sequences. However, the size of damage area increased with an increase in the impact energy at face sheet layers.
- The cracking width increased with the increase in impact energy for all the samples.
- The delamination area which was observed in foam sandwich composite specimens were generally oblong shaped and was as a result of the bending stiffness mismatch between adjacent layers.
- For both types of the stacking sequences, each level of the impact energy had a different shape of the curves, and displacement showed an increase with impact energy.

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